

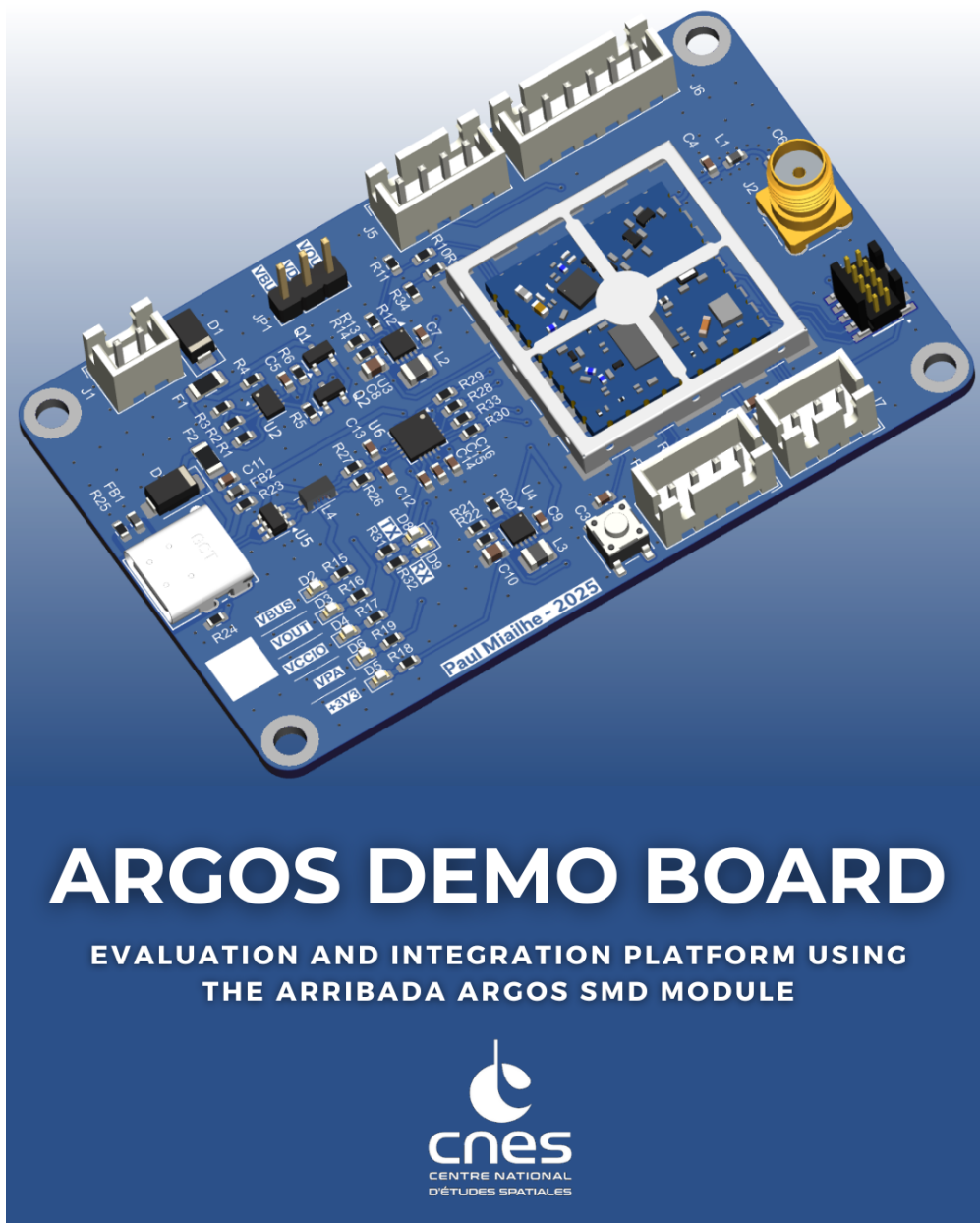
Argos Demo Board Using Arribada SMD Module

Demonstration and evaluation board designed to operate the **Arribada Argos SMD module** via USB and transmit **ARGOS uplink messages to the Kineis satellite constellation**.

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Introduction

This project presents an **Argos Transceiver Demonstration Board** intended for the evaluation, validation, and operation of the **Arribada Argos SMD module**.

The board provides a compact and robust hardware platform to transmit **ARGOS uplink messages to the Kineis satellite constellation** using a standard **USB interface**.

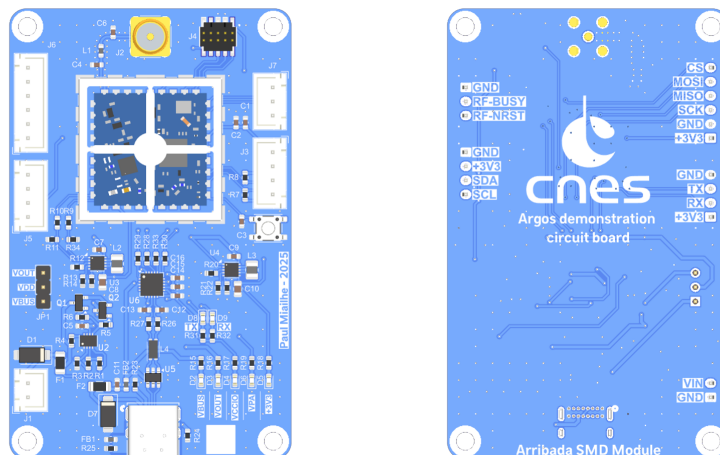
Communication between the host computer and the Arribada SMD module is handled by a **FTDI FT231XQ USB-to-UART bridge**, directly interfacing with the module UART. This approach enables simple control of the transceiver using standard serial terminals or custom software tools.

The demonstration board integrates all essential functions required to operate the Arribada SMD module in a laboratory or development environment, including power supply conditioning, USB-to-UART communication, and RF interfacing. It is primarily intended to support early hardware validation, firmware development, and system-level testing of Argos communication chains prior to integration into a final embedded or avionics system.

This board targets engineers, students, and developers involved in **satellite communications, avionics, IoT, or space-related applications**, and may be used both as a standalone evaluation platform and as a reference hardware design.

Inline with Arribada's open source philosophy, this board deliberately adopts a design distinct from the official demonstration hardware, focusing on simplicity and direct plug-and-play operation from a standard PC. The schematic and architecture are provided for reference, allowing potential users to reuse, adapt, or extend them according to their own needs..

Hardware Design



1. Communication Architecture

Communication between the host PC and the **Arribada Argos SMD module** is performed through the module's **UART interface**.

This interface is used to send **AT commands** from a PC in order to configure the module and transmit **ARGOS uplink messages**. → [See details on GitHub Arribada](#)

The USB-to-UART conversion is handled by a **FTDI FT231XQ USB 2.0 to UART bridge**. This device converts USB data received from a **virtual COM port** into UART signals fully compatible with the Arribada SMD

module.

This architecture allows the board to be operated using:

- Standard serial terminal software
- Custom scripts
- Dedicated test or validation tools

No additional embedded controller is required, which simplifies early testing and debugging.

2. Power Supply Architecture

The card supports two power supply modes to suit different uses.

2.1 Power Source Selection

Power source selection is performed using the **JP1 jumper**, which allows switching between:

- **VBUS → VDD (5V)** supplied by the **USB-C port**
- **VOUT → VDD (5V)** supplied via the **external power input on connector J1**

2.2 External Power Protection

The external power input on **connector J1** is protected by:

- A **resettable PTC fuse 1206L050YR**
- A **LTC4365 protection circuit**

The **LTC4365** provides:

- **Reverse polarity protection**
- **Undervoltage protection** with a threshold set to **2.5 V**
- **Overvoltage protection** for input voltages exceeding **5 V**

2.3 Power Regulation

After the protection stage, the power rail is distributed to two **TPS63900 buck-boost converters**:

- One converter supplies the **RF power amplifier (PA)** and is **enabled and controlled directly by the Arribada SMD module**
 - The second converter is **always enabled** and provides the regulated **3.3 V supply** required for the **Arribada SMD module logic**
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3. Connector Types

The Argos Demo Board uses the following connector families:

- **USB4085-GF-A (GCT)**
USB Type-C receptacle (16-pin) used for:

- USB 2.0 communication with the host PC
- Board power supply via **VBUS**

- **JST B2B-XH series**

Through-hole wire-to-board connectors used for:

- External power input (**2.5 V to 5 V**)
- Digital interfaces (**UART, SPI, GPIO – 3.3 V logic**)
- General-purpose I/O connections

- **FTSH-105-01-L-DV-K (Samtec)**

10-pin JTAG connector used for:

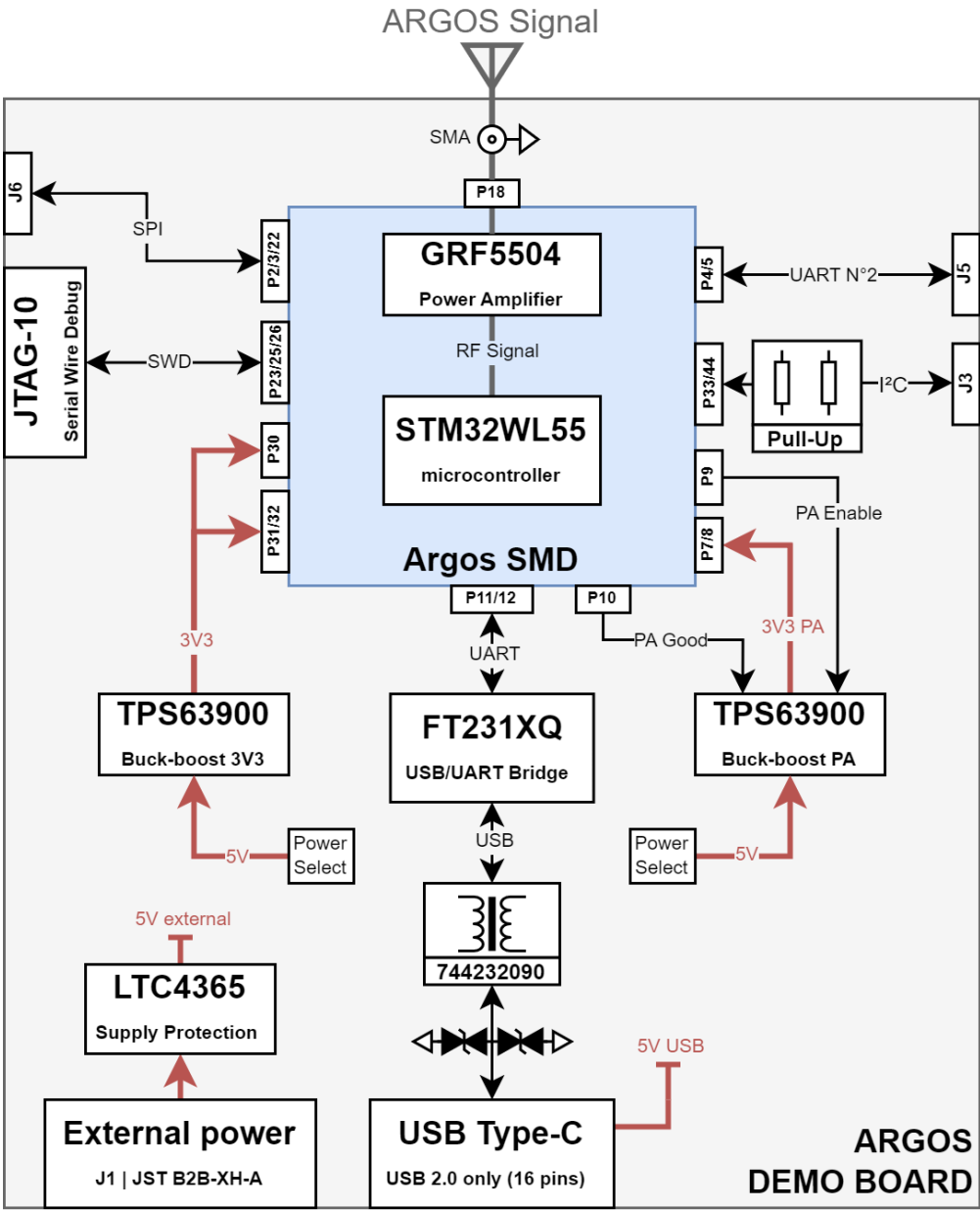
- Programming
- Low-level debugging of the **Arribada Argos SMD module**
- Access to **JTAG/SWD signals** of the embedded **STM32WL55**

- **60312002114503 (Würth Elektronik)**

SMA 50 Ω RF connector used for:

- RF output connection
- External antenna interface for the **ARGOS frequency band**
- ⚠ When transmitting, it is important to have at least a 50 Ohm load connected to the antenna output to protect against impedance mismatch.

4. Functional Diagram



Summary of the Bill of Materials (BOM)

Components used on the board, with their official datasheets:

Reference	Part Number	Role / Function	Datasheet
D1, D7	SMAJ15CA	TVS diode for surge and ESD protection	→ Datasheet Littelfuse
F1, F2	1206L035/16YR	Resettable PTC fuse (overcurrent protection)	→ Datasheet Littelfuse
FB1, FB2	BLM18KG601SN1D	Ferrite bead for EMI suppression	→ Datasheet Murata
L4	744232090	Common-mode choke for USB line filtering	→ Datasheet Würth Elektronik
Q1, Q2	BSN20	N-channel MOSFET for switching and protection	→ Datasheet DIODES INC.
SW1	PTS647	Momentary tactile push button switch	→ Datasheet C&K

Reference	Part Number	Role / Function	Datasheet
U2	LTC4365DDB	Overvoltage, undervoltage and reverse polarity protection IC	→ Datasheet Analog Devices
U3, U4	TPS63900	Buck-boost DC/DC converters	→ Datasheet Texas Instruments
U5	82400102	Low-capacitance TVS diode array for USB protection	→ Datasheet Würth Elektronik
U6	FT231XQ	USB-to-UART bridge	→ Datasheet FTDI
H4	3610213126S	RF shielding cabinet	→ Datasheet Würth Elektronik
J8	USB4085-GF-A	USB 2.0-only 16P Type-C Receptacle connector	→ Datasheet GCT
J2	60312002114503	Coaxial connector SMA	→ Datasheet Würth Elektronik
J4	FTSH-105-01-L-DV-K	JTAG 10-PINS	→ Datasheet Samtec
J1, J3, J5, J6, J7	B2B-XH	Through-hole wire-to-board connectors	→ Datasheet JST
U1	Argos SMD module	Transceiver Argos SMD for uplink Argos satellite	→ Datasheet Arribada

Getting Started

1. Connect the demonstration board to a host computer using a USB cable.
2. Install the **FTDI drivers** if required.
3. Select the desired **power input** using the **JP1 jumper**.
4. Connect an **antenna or a 50 Ω RF load** to the **SMA connector** before powering the board.
5. Open a **serial terminal** or a **custom software tool** on the host computer.
6. Configure the serial port according to the **Arribada Argos SMD module specifications**.
7. Send **AT commands** to configure the module and transmit **ARGOS uplink messages**.
8. If the board is operating correctly, the **5 V and 3.3 V LEDs** should be ON, and **TX/RX activity LEDs** should indicate serial communication.

Refer to the Arribada SMD module documentation for command syntax and operational details.

Regulatory and Operational Notice

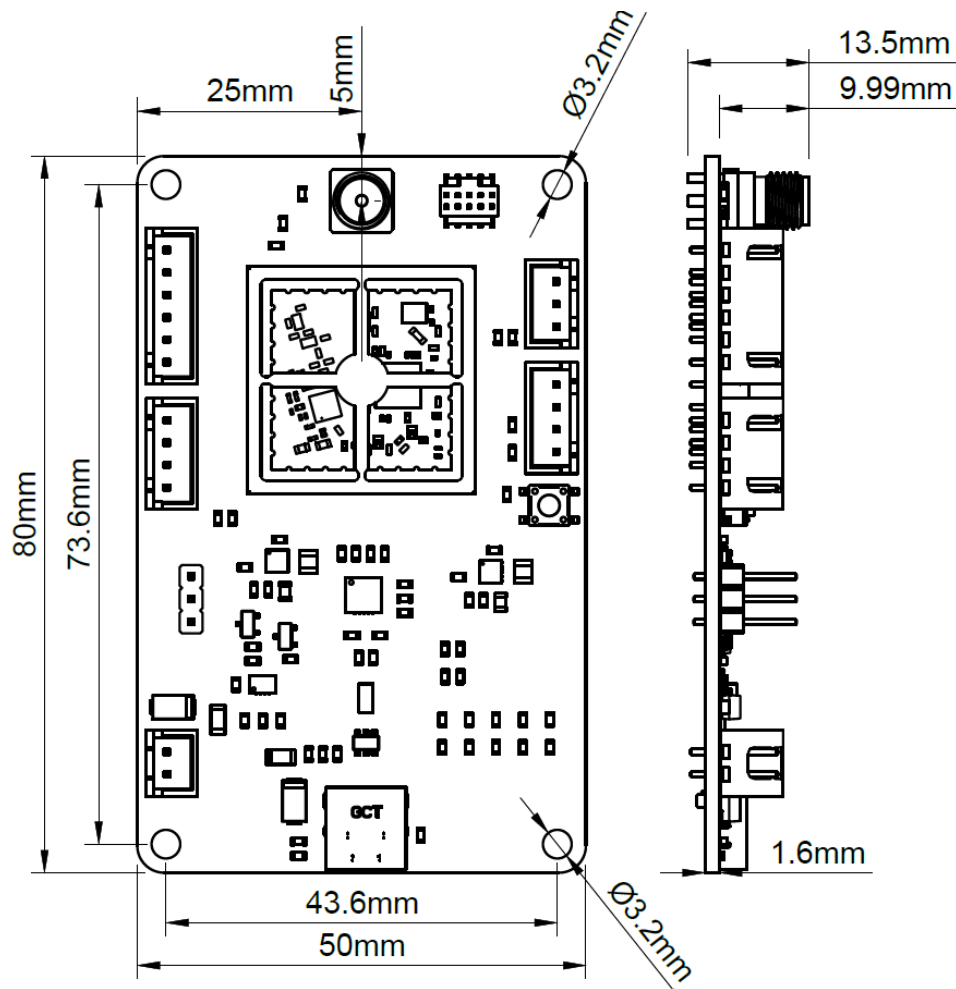
ARGOS transmissions are subject to regulatory constraints and operational authorization.

- Transmission over the ARGOS / Kineis constellation may require prior approval
- Users are responsible for ensuring compliance with applicable regulations

- This board does not grant any transmission rights by itself

Mechanical Characteristics

- Dimensions: **80 × 50 mm**
- Maximum thickness: **13.5 mm**
- Mounting holes: 4 × Ø3.2 mm
- Weight without bracket: ** g**
- **STEP 3D** file available here:



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References & Resources

- **Arribada Initiative** – Official site
See link: <https://arribada.org>

- **Official Argos SMD Arduino Demo** – by Arribada

The full Arduino-based demonstration for the Argos SMD module is available on GitHub.

See link: <https://github.com/arribada/argos-smd-test-arduino>

- **Official Breakout Board FeatherWings** – by Arribada

This breakout board can be used with a Feather board (from Adafruit) to connect the ARGOS-SMD module.

It exposes the ARGOS-SMD module pins to the Feather board and includes a dedicated LDO to power the RF amplifier.

See link: <https://github.com/arribada/featherwings-argos-smd-hw>

- **Official Argos SMD Hardware Module** – by Arribada

The Arribada Argos SMD module is a compact **20 × 20 mm PCB** designed to operate as a standalone module or co-processor, enabling satellite communication via the **Kineis network**.

See link: <https://github.com/arribada/argos-smd-hw>